Lab – File System Commands

Introduction

In this lab, you will use CLI commands to manage files and folders in Windows.

Recommended Equipment

- A computer running Windows

Instructions

Step 1: Access the Windows command prompt.

a. Log on to a Windows computer. The account **ITEUser** is used as the example user account throughout this lab.

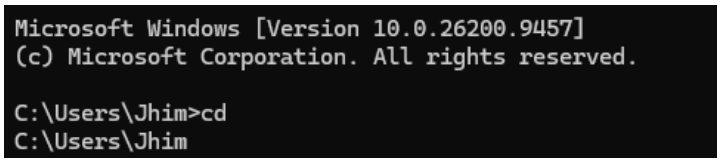
b. To access the Windows command prompt, click **Start** and enter **cmd**. Select **Command Prompt** or **cmd.exe** to continue.

Step 2: Create and change directories.

In this step, you will use the change directory (**cd**), make directory (**md**), and directory (**dir**) commands.

Note: A directory is another word for folder. Directory and folder are used interchangeably throughout this lab.

a. Type **cd** at the command prompt.



Question:

What is the current directory?

The current directory is C:\Users\Jhim.

b. Type **dir** at the command prompt to list the files and folders that are in the current folder.

```
C:\Users\ITEUser> dir
 Volume in drive C has no label.
 Volume Serial Number is 9055-35E9

 Directory of C:\Users\ITEUser

04/27/2019  09:21 AM    <DIR>          .
04/27/2019  09:21 AM    <DIR>          ..
04/25/2019  11:39 AM    <DIR>          3D Objects
04/25/2019  11:39 AM    <DIR>          Contacts
04/26/2019  10:29 AM    <DIR>          Desktop
04/25/2019  11:39 AM    <DIR>          Documents
04/25/2019  11:39 AM    <DIR>          Downloads
<some output omitted>
```

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```
               0 File(s)              0 bytes
              15 Dir(s)  32,671,969,280 bytes free
```

```
C:\Users\Jhim>dir
 Volume in drive C is OS
 Volume Serial Number is A448-0424

 Directory of C:\Users\Jhim

06/10/2026  06:57 pm    <DIR>          .
30/01/2025  07:31 pm    <DIR>          ..
30/11/2024  01:25 pm    <DIR>          .android
08/01/2024  10:14 am    <DIR>          .arduinoIDE
05/02/2024  08:07 am    <DIR>          .idlerc
25/01/2026  11:18 pm    <DIR>          .mgltools
01/09/2026  11:43 am               174 .packettracer
23/02/2024  10:14 am    <DIR>          3D Objects
01/09/2026  01:07 pm    <DIR>          Cisco Packet Tracer 9.0.1
30/01/2025  07:38 pm    <DIR>          Contacts
04/11/2025  01:59 pm    <DIR>          Documents
06/10/2026  07:00 pm    <DIR>          Downloads
30/01/2025  07:38 pm    <DIR>          Favorites
30/01/2025  07:38 pm    <DIR>          Links
30/01/2025  07:38 pm    <DIR>          Music
06/10/2026  06:50 pm    <DIR>          OneDrive
08/04/2022  08:45 am    <DIR>          Pictures
30/01/2025  07:38 pm    <DIR>          Saved Games
30/01/2025  07:38 pm    <DIR>          Searches
18/03/2022  11:00 am    <DIR>          Tracing
13/07/2026  12:53 am    <DIR>          Videos
15/07/2026  10:08 pm    <DIR>          WPS Cloud
22/04/2026  10:36 pm    <DIR>          WPSDrive
               1 File(s)            174 bytes
              22 Dir(s)  88,468,176,896 bytes free
```

c. In the current directory, use the **md** command to create three new folders: **ITEfolder1**, **ITEfolder2**, and **ITEfolder3**. Type **md ITEfolder1** and press **Enter**. Create **ITEfolder2** and **ITEfolder3**. (**Note**: The command **mkdir** performs the same function as the command **md**.)

```
C:\Users\ITEUser> md ITEfolder1
C:\Users\ITEUser> md ITEfolder2 ITEfolder3
```

```
C:\Users\Jhim>md ITEfolder1

C:\Users\Jhim>md ITEfolder2 ITEfolder3
```

d. Type **dir** to verify that the folders have been created.

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```
C:\Users\Jhim>dir
 Volume in drive C is OS
 Volume Serial Number is A448-0424

 Directory of C:\Users\Jhim

06/10/2026  07:09 pm    <DIR>          .
30/01/2025  07:31 pm    <DIR>          ..
30/11/2024  01:25 pm    <DIR>          .android
08/01/2024  10:14 am    <DIR>          .arduinoIDE
05/02/2024  08:07 am    <DIR>          .idlerc
25/01/2026  11:18 pm    <DIR>          .mgltools
01/09/2026  11:43 am               174 .packettracer
23/02/2024  10:14 am    <DIR>          3D Objects
01/09/2026  01:07 pm    <DIR>          Cisco Packet Tracer 9.0.1
30/01/2025  07:38 pm    <DIR>          Contacts
04/11/2025  01:59 pm    <DIR>          Documents
06/10/2026  07:07 pm    <DIR>          Downloads
30/01/2025  07:38 pm    <DIR>          Favorites
06/10/2026  07:08 pm    <DIR>          ITEfolder1
06/10/2026  07:09 pm    <DIR>          ITEfolder2
06/10/2026  07:09 pm    <DIR>          ITEfolder3
30/01/2025  07:38 pm    <DIR>          Links
30/01/2025  07:38 pm    <DIR>          Music
06/10/2026  06:50 pm    <DIR>          OneDrive
08/04/2022  08:45 am    <DIR>          Pictures
30/01/2025  07:38 pm    <DIR>          Saved Games
30/01/2025  07:38 pm    <DIR>          Searches
18/03/2022  11:00 am    <DIR>          Tracing
13/07/2026  12:53 am    <DIR>          Videos
15/07/2026  10:08 pm    <DIR>          WPS Cloud
22/04/2026  10:36 pm    <DIR>          WPSDrive
               1 File(s)            174 bytes
              25 Dir(s)  88,469,188,608 bytes free
```

e. Type **cd ITEfolder3** at the command prompt and press **Enter**.

```
C:\Users\Jhim>cd ITEfolder3

C:\Users\Jhim\ITEfolder3>
```

Question:

Which folder are you in now?

I am in ITEfolder3 now

f. Create **ITEfolder4** in the current directory. Within the **ITEfolder4** folder, create a folder named **ITEfolder5**. Use the **dir** command to verify the folder creation.

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```
C:\Users\Jhim\ITEfolder3>md ITEfolder4

C:\Users\Jhim\ITEfolder3>cd ITEfolder4

C:\Users\Jhim\ITEfolder3\ITEfolder4>md ITEfolder5

C:\Users\Jhim\ITEfolder3\ITEfolder4>dir
 Volume in drive C is OS
 Volume Serial Number is A448-0424

 Directory of C:\Users\Jhim\ITEfolder3\ITEfolder4

06/10/2026  07:18 pm    <DIR>          .
06/10/2026  07:16 pm    <DIR>          ..
06/10/2026  07:18 pm    <DIR>          ITEfolder5
               0 File(s)              0 bytes
               3 Dir(s)  88,463,323,136 bytes free
```

Question:

What command or commands did you use to create the nested folders?

The commands used to create the nested folders were md ITEfolder4, cd ITEfolder4, md ITEfolder5 and dir.

g. Change directory as necessary until you are in ITEfolder5.

```
C:\Users\Jhim\ITEfolder3\ITEfolder4>cd ITEfolder5

C:\Users\Jhim\ITEfolder3\ITEfolder4\ITEfolder5>
```

h. Type **cd ..** to change the current directory. Each **..** is a shortcut to move up one level in the directory tree.

```
C:\Users\Jhim\ITEfolder3\ITEfolder4\ITEfolder5>cd ..

C:\Users\Jhim\ITEfolder3\ITEfolder4>cd ..

C:\Users\Jhim\ITEfolder3>cd ..\ITEfolder1
```

Question:

After issuing the **cd ..** command, what is your directory now?

C:\Users\Jhim\ITEfolder3\ITEfolder4

What would be the current directory if you issue this command at C:\Users\ITEUser\ITEfolder3\ITEFolder4?

C:\Users\Jhim\ITEfolder3

Step 3: Create text files.

a. Navigate to the **C:\Users\ITEUser\ITEfolder1** directory. Type **cd ..\ITEfolder1** at the prompt.

```
C:\Users\Jhim\ITEfolder3>cd ..\ITEfolder1

C:\Users\Jhim\ITEfolder1>echo This is doc1
```

b. Type **echo This is doc1.txt > doc1.txt** at the command prompt. The **echo** command is used to display a message at the command prompt. The **>** is used to redirect the message from the screen to a file. For

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example, in the first line, the message **This is doc1.txt** is redirected into a new file named **doc1.txt**. Use the **echo** command and **>** redirect to create these files: **doc2.txt**, **file1.txt**, and **file2.txt**.

```
C:\Users\ITEUser\ITEfolder1> echo This is doc1.txt > doc1.txt
C:\Users\ITEUser\ITEfolder1> echo This is doc2.txt > doc2.txt
C:\Users\ITEUser\ITEfolder1> echo This is file1.txt > file1.txt
C:\Users\ITEUser\ITEfolder1> echo This is file2.txt > file2.txt
```

```
C:\Users\Jhim\ITEfolder1>echo This is doc1.txt > doc1.txt

C:\Users\Jhim\ITEfolder1>echo This is doc2.txt > doc2.txt

C:\Users\Jhim\ITEfolder1>echo This is file1.txt > file1.txt

C:\Users\Jhim\ITEfolder1>echo This is file2.txt > file2.txt
```

c. Use the **dir** command to verify the files are in the **ITEfolder1** folder.

```
C:\Users\ITEUser\ITEFolder1> dir
<some output omitted>
04/29/2019  08:05 AM                19 doc1.txt
04/29/2019  08:06 AM                19 doc2.txt
04/29/2019  08:08 AM                20 file1.txt
04/29/2019  08:08 AM                20 file2.txt
               4 File(s)             78 bytes
               2 Dir(s)  32,625,397,760 bytes free
```

```
C:\Users\Jhim\ITEfolder1>dir
 Volume in drive C is OS
 Volume Serial Number is A448-0424

 Directory of C:\Users\Jhim\ITEfolder1

06/10/2026  07:37 pm    <DIR>          .
06/10/2026  07:09 pm    <DIR>          ..
06/10/2026  07:36 pm                19 doc1.txt
06/10/2026  07:36 pm                19 doc2.txt
06/10/2026  07:36 pm                20 file1.txt
06/10/2026  07:37 pm                20 file2.txt
               4 File(s)             78 bytes
               2 Dir(s)  88,412,487,680 bytes free
```

d. Use the **more** or **type** command to view the content of the newly created text files.

```
C:\Users\ITEUser\ITEfolder1> more doc1.txt
This is doc1.txt

C:\Users\ITEUser\ITEfolder1> type doc2.txt
This is doc2.txt
```

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```
C:\Users\Jhim\ITEfolder1>more doc1.txt
This is doc1.txt

C:\Users\Jhim\ITEfolder1>more doc2.txt
This is doc2.txt
```

Step 4: Copy, delete, and move files.

a. At the command prompt, type **move doc2.txt C:\Users\ITEUser\ITEfolder2** to move the file **doc2.txt** to the **C:\Users\ITEUser\ITEfolder2** directory.

```
C:\Users\ITEUser\ITEfolder1> move doc2.txt C:\Users\ITEUser\ITEfolder2
        1 file(s) moved.
```

```
C:\Users\Jhim\ITEfolder1>move doc2.txt C:\Users\Jhim\ITEfolder2
        1 file(s) moved.
```

b. Type **dir** at the prompt to verify that **doc2.txt** is no longer in the current directory.

```
C:\Users\Jhim\ITEfolder1>dir
 Volume in drive C is OS
 Volume Serial Number is A448-0424

 Directory of C:\Users\Jhim\ITEfolder1

06/10/2026  07:41 pm    <DIR>          .
06/10/2026  07:09 pm    <DIR>          ..
06/10/2026  07:36 pm                19 doc1.txt
06/10/2026  07:36 pm                20 file1.txt
06/10/2026  07:37 pm                20 file2.txt
               3 File(s)             59 bytes
               2 Dir(s)  88,403,263,488 bytes free
```

c. Navigate to **C:\Users\ITEUser\ITEfolder2** to change the directory to **ITEfolder2**. Type **dir** at the prompt to verify **doc2.txt** has been moved.

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```
C:\Users\Jhim\ITEfolder1>cd C:\Users\Jhim\ITEfolder2

C:\Users\Jhim\ITEfolder2>dir
 Volume in drive C is OS
 Volume Serial Number is A448-0424

 Directory of C:\Users\Jhim\ITEfolder2

06/10/2026  07:41 pm    <DIR>          .
06/10/2026  07:09 pm    <DIR>          ..
06/10/2026  07:36 pm                19 doc2.txt
               1 File(s)             19 bytes
               2 Dir(s)  88,400,154,624 bytes free
```

d. Type **copy doc2.txt doc2_copy.txt** to create a copy of **doc2.txt**. Type **dir** at the prompt to verify a copy of the file has been created.

```
C:\Users\ITEUser\ITEfolder2> dir
<some output omitted>
Directory of C:\Users\ITEUser\ITEfolder2

04/30/2019  09:07 AM                19 doc2.txt
04/30/2019  09:07 AM                19 doc2_copy.txt
               2 File(s)             38 bytes
               2 Dir(s)  31,753,068,544 bytes free
```

```
C:\Users\Jhim\ITEfolder2>copy doc2.txt doc2_copy.txt
        1 file(s) copied.

C:\Users\Jhim\ITEfolder2>dir
 Volume in drive C is OS
 Volume Serial Number is A448-0424

 Directory of C:\Users\Jhim\ITEfolder2

06/10/2026  07:43 pm    <DIR>          .
06/10/2026  07:09 pm    <DIR>          ..
06/10/2026  07:36 pm                19 doc2.txt
06/10/2026  07:36 pm                19 doc2_copy.txt
               2 File(s)             38 bytes
               2 Dir(s)  88,395,051,008 bytes free
```

e. Use the **move** command to move **doc2_copy.txt** to **ITEfolder1**. Type **move doc2_copy.txt ..\ITEfolder1**.

```
C:\Users\ITEUser\ITEfolder2> move doc2_copy.txt ..\ITEfolder1
        1 file(s) moved.
```

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```
C:\Users\Jhim\ITEfolder2>move doc2_copy.txt ..\ITEfolder1
        1 file(s) moved.
```

f. A copy of the file **doc2.txt** can be created in a new location using the **copy** command. At the prompt, enter the **copy doc2.txt ..\ITEfolder1\doc2_new.txt**.

```
C:\Users\ITEUser\ITEfolder2> copy doc2.txt ..\ITEfolder1\doc2_new.txt
        1 file(s) copied.
```

```
C:\Users\Jhim\ITEfolder2>copy doc2.txt ..\ITEfolder1\doc2_new.txt
        1 file(s) copied.
```

g. The file **doc2.txt** can also be moved to a new location with a new filename using the **move** command. Type **move doc2.txt ..\ITEfolder1\doc2_new.txt** at the prompt.

```
C:\Users\ITEUser\ITEfolder2> move doc2.txt ..\ITEfolder1\doc2_move.txt
        1 file(s) moved.
```

```
C:\Users\Jhim\ITEfolder2>move doc2.txt ..\ITEfolder1\doc2_move.txt
        1 file(s) moved.
```

h. Type **dir ..\ITEfolder1** to view the content in **ITEfolder1** without leaving the current directory.

```
C:\Users\ITEUser\ITEfolder2> dir ..\ITEfolder1
<some output omitted>
 Directory of C:\Users\ITEUser\ITEfolder1

04/29/2019  12:08 PM    <DIR>          .
04/29/2019  12:08 PM    <DIR>          ..
04/29/2019  08:05 AM                19 doc1.txt
04/29/2019  08:06 AM                19 doc2_copy.txt
04/29/2019  08:06 AM                19 doc2_move.txt
04/29/2019  08:06 AM                19 doc2_new.txt
04/29/2019  08:08 AM                20 file1.txt
04/29/2019  08:08 AM                20 file2.txt
               6 File(s)            116 bytes
               2 Dir(s)  31,467,700,224 bytes free
```

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```
C:\Users\Jhim\ITEfolder2>dir ..\ITEfolder1
 Volume in drive C is OS
 Volume Serial Number is A448-0424

 Directory of C:\Users\Jhim\ITEfolder1

06/10/2026  07:47 pm    <DIR>          .
06/10/2026  07:09 pm    <DIR>          ..
06/10/2026  07:36 pm                19 doc1.txt
06/10/2026  07:36 pm                19 doc2_copy.txt
06/10/2026  07:36 pm                19 doc2_move.txt
06/10/2026  07:36 pm                19 doc2_new.txt
06/10/2026  07:36 pm                20 file1.txt
06/10/2026  07:37 pm                20 file2.txt
               6 File(s)            116 bytes
               2 Dir(s)  88,404,643,840 bytes free
```

i. Change the current directory to **ITEfolder1**. Type **cd ..\ITEfolder1** at the prompt.

```
C:\Users\Jhim\ITEfolder2>cd ..\ITEfolder1
```

j. Move **file1.txt** and **file2.txt** into **ITEfolder3**. To move all the files that contain the word **file** into **ITEfolder3** with one command, use a **wildcard** (*) character to represent one or more characters. Type **move file* ..\ITEfolder3**.

```
C:\Users\ITEUser\ITEfolder1> move file* ..\ITEfolder3
C:\Users\ITEUser\ITEfolder1\file1.txt
C:\Users\ITEUser\ITEfolder1\file2.txt
        2 file(s) moved.
```

```
C:\Users\Jhim\ITEfolder1>move file* ..\ITEfolder3
C:\Users\Jhim\ITEfolder1\file1.txt
C:\Users\Jhim\ITEfolder1\file2.txt
        2 file(s) moved.
```

k. To view the content of the file doc2_new.txt, use the **type** or **more** command at the prompt.

```
C:\Users\ITEUser\ITEfolder1> type doc2_new.txt
This is doc2.txt
```

```
C:\Users\Jhim\ITEfolder1>type doc2_new.txt
This is doc2.txt
```

l. Use the **ren** command to rename doc2_new.txt to doc3.txt. Use the **dir** command to display the content in the directory.

```
c:\Users\ITEUser\ITEfolder1> ren doc2_new.txt file.log
```

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```
C:\Users\Jhim\ITEfolder1>ren doc2_new.txt file.log

C:\Users\Jhim\ITEfolder1>dir
 Volume in drive C is OS
 Volume Serial Number is A448-0424

 Directory of C:\Users\Jhim\ITEfolder1

06/10/2026  07:50 pm    <DIR>          .
06/10/2026  07:09 pm    <DIR>          ..
06/10/2026  07:36 pm                19 doc1.txt
06/10/2026  07:36 pm                19 doc2_copy.txt
06/10/2026  07:36 pm                19 doc2_move.txt
06/10/2026  07:36 pm                19 file.log
               4 File(s)             76 bytes
               2 Dir(s)  88,403,628,032 bytes free
```

m. Use the **type** or **more** command to view the content of the newly rename file **file.log.**

```
C:\Users\ITEUser\ITEfolder1> more file.log
This is doc2.txt
```

```
C:\Users\Jhim\ITEfolder1>more file.log
This is doc2.txt
```

n. To delete a file or multiple files, use the **del** command with the file names at the prompt.

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```
C:\Users\Jhim\ITEfolder1>del *doc2*

C:\Users\Jhim\ITEfolder1>dir
 Volume in drive C is OS
 Volume Serial Number is A448-0424

 Directory of C:\Users\Jhim\ITEfolder1

06/10/2026  08:07 pm    <DIR>          .
06/10/2026  07:09 pm    <DIR>          ..
06/10/2026  07:36 pm                19 doc1.txt
06/10/2026  07:36 pm                19 file.log
               2 File(s)             38 bytes
               2 Dir(s)  88,423,649,280 bytes free

C:\Users\Jhim\ITEfolder1>del *.*
C:\Users\Jhim\ITEfolder1\*.*, Are you sure (Y/N)? Y

C:\Users\Jhim\ITEfolder1>dir
 Volume in drive C is OS
 Volume Serial Number is A448-0424

 Directory of C:\Users\Jhim\ITEfolder1

06/10/2026  08:07 pm    <DIR>          .
06/10/2026  07:09 pm    <DIR>          ..
               0 File(s)              0 bytes
               2 Dir(s)  88,424,525,824 bytes free
```

Question:

What single command would you use to delete all the files with doc2 in the filename? Use the command to remove the files with doc2 in the filename.

del *doc2*

What command would you use to delete all the files in the directory? Use the command to remove the files.

del *.*

Step 5: Use the xcopy and robocopy commands.

In this step, the **xcopy** and **robocopy** commands are used to copy the content in a directory.

a. View the content of ITEfolder3.

```
C:\Users\ITEUser\ITEfolder1> dir ..\ITEfolder3
<some output omitted>
 Directory of c:\Users\ITEUser\ITEfolder3

04/29/2019  03:11 PM    <DIR>          .
```

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```
04/29/2019  03:11 PM    <DIR>          ..
04/29/2019  03:05 PM                20 file1.txt
04/29/2019  03:05 PM                20 file2.txt
04/29/2019  03:01 PM    <DIR>          ITEfolder4
               2 File(s)             40 bytes
               3 Dir(s)  31,492,157,440 bytes free
```

```
C:\Users\Jhim\ITEfolder1>dir ..\ITEfolder3
 Volume in drive C is OS
 Volume Serial Number is A448-0424

 Directory of C:\Users\Jhim\ITEfolder3

06/10/2026  07:50 pm    <DIR>          .
06/10/2026  07:09 pm    <DIR>          ..
06/10/2026  07:36 pm                20 file1.txt
06/10/2026  07:37 pm                20 file2.txt
06/10/2026  07:18 pm    <DIR>          ITEfolder4
               2 File(s)             40 bytes
               3 Dir(s)  88,421,048,320 bytes free
```

b. Type **xcopy ..\ITEfolder3 .** at the prompt to copy the content of **ITEfolder3** to **ITEfolder1**. Note the **.** at the end of the command. It is a shortcut for the current directory.

```
C:\Users\ITEUser\ITEfolder1> xcopy ..\ITEfolder3 .
..\ITEfolder3\file1.txt
..\ITEfolder3\file2.txt
2 File(s) copied
```

```
C:\Users\Jhim\ITEfolder1>xcopy ..\ITEfolder3 .
..\ITEfolder3\file1.txt
..\ITEfolder3\file2.txt
2 File(s) copied
```

c. At the prompt, type **dir** to display the content of **ITEfolder1**. Only the files in the **ITEfolder3** were copied into **ITEfolder1**. The directory **ITEfolder4** was not copied into **ITEfolder3**.

```
C:\Users\ITEUser\ITEfolder1> dir
<some output omitted>
 Directory of C:\Users\ITEUser\ITEfolder1
04/29/2019  03:16 PM    <DIR>          .
04/29/2019  03:16 PM    <DIR>          ..
04/29/2019  03:05 PM                20 file1.txt
04/29/2019  03:05 PM                20 file2.txt
               2 File(s)             40 bytes
               2 Dir(s)  31,491,321,856 bytes free
```

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```
C:\Users\Jhim\ITEfolder1>dir
 Volume in drive C is OS
 Volume Serial Number is A448-0424

 Directory of C:\Users\Jhim\ITEfolder1

06/10/2026  08:14 pm    <DIR>          .
06/10/2026  07:09 pm    <DIR>          ..
06/10/2026  07:36 pm                20 file1.txt
06/10/2026  07:37 pm                20 file2.txt
               2 File(s)             40 bytes
               2 Dir(s)  88,421,367,808 bytes free
```

d. Use **help xcopy** or **xcopy /?** to determine which switch would allow the **xcopy** command to copy **all** the files and directories.

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```
C:\Users\Jhim\ITEfolder1>help xcopy
Copies files and directory trees.

XCOPY source [destination] [/A | /M] [/D[:date]] [/P] [/S [/E]] [/V] [/W]
                           [/C] [/I] [/-I] [/Q] [/F] [/L] [/G] [/H] [/R] [/T]
                           [/U] [/K] [/N] [/O] [/X] [/Y] [/-Y] [/Z] [/B] [/J]
                           [/EXCLUDE:file1[+file2][+file3]...] [/COMPRESS]
                           [/[-]SPARSE] [/NOCLONE]

  source       Specifies the file(s) to copy.
  destination  Specifies the location and/or name of new files.
  /A           Copies only files with the archive attribute set,
               doesn't change the attribute.
  /M           Copies only files with the archive attribute set,
               turns off the archive attribute.
  /D:m-d-y     Copies files changed on or after the specified date.
               If no date is given, copies only those files whose
               source time is newer than the destination time.
  /EXCLUDE:file1[+file2][+file3]...
               Specifies a list of files containing strings.  Each string
               should be in a separate line in the files.  When any of the
               strings match any part of the absolute path of the file to be
               copied, that file will be excluded from being copied.  For
               example, specifying a string like \obj\ or .obj will exclude
               all files underneath the directory obj or all files with the
               .obj extension respectively.
  /P           Prompts you before creating each destination file.
  /S           Copies directories and subdirectories except empty ones.
  /E           Copies directories and subdirectories, including empty ones.
               Same as /S /E. May be used to modify /T.
  /V           Verifies the size of each new file.
  /W           Prompts you to press a key before copying.
  /C           Continues copying even if errors occur.
  /I           If destination does not exist and copying more than one file,
               assumes that destination must be a directory.
  /-I          If destination does not exist and copying a single specified file,
               assumes that destination must be a file.
  /Q           Does not display file names while copying.
  /F           Displays full source and destination file names while copying.
  /L           Displays files that would be copied.
  /G           Allows the copying of encrypted files to destination that does
               not support encryption.
  /H           Copies hidden and system files also.
  /R           Overwrites read-only files.
  /T           Creates directory structure, but does not copy files. Does not
               include empty directories or subdirectories. /T /E includes
               empty directories and subdirectories.
  /U           Copies only files that already exist in destination.
  /K           Copies attributes. Normal Xcopy will reset read-only attributes.
  /N           Copies using the generated short names.
  /O           Copies file ownership and ACL information.
  /X           Copies file audit settings (implies /O).
  /Y           Suppresses prompting to confirm you want to overwrite an
               existing destination file.
  /-Y          Causes prompting to confirm you want to overwrite an
               existing destination file.
  /Z           Copies networked files in restartable mode.
  /B           Copies the Symbolic Link itself versus the target of the link.
  /J           Copies using unbuffered I/O. Recommended for very large files.
  /COMPRESS    Request network compression during file transfer where
               applicable.
  /[-]SPARSE   Enable or disable retaining the sparse state of files during
               copy. /-SPARSE overrides /SPARSE if both are provided.
  /NOCLONE     Do not attempt block cloning as an optimization.

The switch /Y may be preset in the COPYCMD environment variable.
This may be overridden with /-Y on the command line.
```

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```
C:\Users\Jhim\ITEfolder1>xcopy /?
Copies files and directory trees.

XCOPY source [destination] [/A | /M] [/D[:date]] [/P] [/S [/E]] [/V] [/W]
                           [/C] [/I] [/-I] [/Q] [/F] [/L] [/G] [/H] [/R] [/T]
                           [/U] [/K] [/N] [/O] [/X] [/Y] [/-Y] [/Z] [/B] [/J]
                           [/EXCLUDE:file1[+file2][+file3]...] [/COMPRESS]
                           [/[-]SPARSE] [/NOCLONE]

  source       Specifies the file(s) to copy.
  destination  Specifies the location and/or name of new files.
  /A           Copies only files with the archive attribute set,
               doesn't change the attribute.
  /M           Copies only files with the archive attribute set,
               turns off the archive attribute.
  /D:m-d-y     Copies files changed on or after the specified date.
               If no date is given, copies only those files whose
               source time is newer than the destination time.
  /EXCLUDE:file1[+file2][+file3]...
               Specifies a list of files containing strings.  Each string
               should be in a separate line in the files.  When any of the
               strings match any part of the absolute path of the file to be
               copied, that file will be excluded from being copied.  For
               example, specifying a string like \obj\ or .obj will exclude
               all files underneath the directory obj or all files with the
               .obj extension respectively.
  /P           Prompts you before creating each destination file.
  /S           Copies directories and subdirectories except empty ones.
  /E           Copies directories and subdirectories, including empty ones.
               Same as /S /E. May be used to modify /T.
  /V           Verifies the size of each new file.
  /W           Prompts you to press a key before copying.
  /C           Continues copying even if errors occur.
  /I           If destination does not exist and copying more than one file,
               assumes that destination must be a directory.
  /-I          If destination does not exist and copying a single specified file,
               assumes that destination must be a file.
  /Q           Does not display file names while copying.
  /F           Displays full source and destination file names while copying.
  /L           Displays files that would be copied.
  /G           Allows the copying of encrypted files to destination that does
               not support encryption.
  /H           Copies hidden and system files also.
  /R           Overwrites read-only files.
  /T           Creates directory structure, but does not copy files. Does not
               include empty directories or subdirectories. /T /E includes
               empty directories and subdirectories.
  /U           Copies only files that already exist in destination.
  /K           Copies attributes. Normal Xcopy will reset read-only attributes.
  /N           Copies using the generated short names.
  /O           Copies file ownership and ACL information.
  /X           Copies file audit settings (implies /O).
  /Y           Suppresses prompting to confirm you want to overwrite an
               existing destination file.
  /-Y          Causes prompting to confirm you want to overwrite an
               existing destination file.
  /Z           Copies networked files in restartable mode.
  /B           Copies the Symbolic Link itself versus the target of the link.
  /J           Copies using unbuffered I/O. Recommended for very large files.
  /COMPRESS    Request network compression during file transfer where
               applicable.
  /[-]SPARSE   Enable or disable retaining the sparse state of files during
               copy. /-SPARSE overrides /SPARSE if both are provided.
  /NOCLONE     Do not attempt block cloning as an optimization.

The switch /Y may be preset in the COPYCMD environment variable.
This may be overridden with /-Y on the command line.
```

Question:

What option allows you to copy all the files and directories, including the empty directories?

xcopy /? allows you to copy all the files and directories, including the empty directories.

e. Because **ITEfolder4** is a subfolder and **ITEfolder5** is both a subfolder and an empty folder, **/E** is needed to copy all the contents of **ITEfolder3** and the empty subfolder.

Type **xcopy /E ..\ITEfolder3 .** at the prompt to copy the files. When prompted, type **a** to allow overwriting the existing files.

```
c:\Users\ITEUser\ITEfolder1> xcopy /E ..\ITEfolder3
Overwrite C:\Users\ITEUser\ITEfolder1\file1.txt (Yes/No/All)? a
..\ITEfolder3\file1.txt
..\ITEfolder3\file2.txt
2 File(s) copied
```

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```
C:\Users\Jhim\ITEfolder1>xcopy /E ..\ITEfolder3
Overwrite C:\Users\Jhim\ITEfolder1\file1.txt (Yes/No/All)? a
..\ITEfolder3\file1.txt
..\ITEfolder3\file2.txt
2 File(s) copied
```

f. Verify that the **ITEfolder4** and **ITEfolder5** were also copied in **ITEfolder1**.

```
c:\Users\ITEUser\ITEfolder1> dir
<some output omitted>
 Directory of c:\Users\ITEUser\ITEfolder1

04/29/2019  04:41 PM    <DIR>          .
04/29/2019  04:41 PM    <DIR>          ..
04/29/2019  03:05 PM                20 file1.txt
04/29/2019  03:05 PM                20 file2.txt
04/29/2019  04:41 PM    <DIR>          ITEfolder4
               2 File(s)             40 bytes
               3 Dir(s)  31,493,193,728 bytes free

c:\Users\ITEUser\ITEfolder1> dir ITEfolder4
<some output omitted>
 Directory of c:\Users\ITEUser\ITEfolder1\ITEfolder4

04/29/2019  04:41 PM    <DIR>          .
04/29/2019  04:41 PM    <DIR>          ..
04/29/2019  03:00 PM    <DIR>          ITEfolder5
               0 File(s)              0 bytes
               3 Dir(s)  31,493,193,728 bytes free
```

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```
C:\Users\Jhim\ITEfolder1>dir
 Volume in drive C is OS
 Volume Serial Number is A448-0424

 Directory of C:\Users\Jhim\ITEfolder1

06/10/2026  08:20 pm    <DIR>          .
06/10/2026  07:09 pm    <DIR>          ..
06/10/2026  07:36 pm                20 file1.txt
06/10/2026  07:37 pm                20 file2.txt
06/10/2026  08:20 pm    <DIR>          ITEfolder4
               2 File(s)             40 bytes
               3 Dir(s)  88,415,559,680 bytes free

C:\Users\Jhim\ITEfolder1>dir ITEfolder4
 Volume in drive C is OS
 Volume Serial Number is A448-0424

 Directory of C:\Users\Jhim\ITEfolder1\ITEfolder4

06/10/2026  08:20 pm    <DIR>          .
06/10/2026  08:20 pm    <DIR>          ..
06/10/2026  07:18 pm    <DIR>          ITEfolder5
               0 File(s)              0 bytes
               3 Dir(s)  88,413,716,480 bytes free
```

g. The robocopy command can also be used to copy the content of a directory to a new destination. The robocopy command has more capabilities than the built-in Windows copy and xcopy command, such as resume copying after a network interruption, skip files that appear to be identical to the files in the destination folders, and mirror a directory by keeping the destination directory in sync with the source directory.

Copy the ITEfolder4 content to ITEUser folder using the robocopy command.

```
C:\Users\ITEUser\ITEfolder1> robocopy /E
C:\Users\ITEUser\ITEfolder3\ITEfolder4\ C:\Users\ITEUser
```

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```
C:\Users\Jhim\ITEfolder1>robocopy /E C:\Users\Jhim\ITEfolder3\ITEfolder4\ C:\Users\Jhim

------------------------------------------------------------------------------
   ROBOCOPY     ::     Robust File Copy for Windows
------------------------------------------------------------------------------

  Started : Tuesday, 6 October 2026 8:23:19 pm
   Source : C:\Users\Jhim\ITEfolder3\ITEfolder4\
     Dest : C:\Users\Jhim\

    Files : *.*

  Options : *.* /S /E /DCOPY:DA /COPY:DAT /R:1000000 /W:30

------------------------------------------------------------------------------

                           0    C:\Users\Jhim\ITEfolder3\ITEfolder4\
        *EXTRA Dir        -1    C:\Users\Jhim\.android\
        *EXTRA Dir        -1    C:\Users\Jhim\.arduinoIDE\
        *EXTRA Dir        -1    C:\Users\Jhim\.idlerc\
        *EXTRA Dir        -1    C:\Users\Jhim\.mgltools\
        *EXTRA Dir        -1    C:\Users\Jhim\3D Objects\
        *EXTRA Dir        -1    C:\Users\Jhim\AppData\
        *EXTRA Dir        -1    C:\Users\Jhim\Application Data\
        *EXTRA Dir        -1    C:\Users\Jhim\Cisco Packet Tracer 9.0.1\
        *EXTRA Dir        -1    C:\Users\Jhim\Contacts\
        *EXTRA Dir        -1    C:\Users\Jhim\Cookies\
        *EXTRA Dir        -1    C:\Users\Jhim\Documents\
        *EXTRA Dir        -1    C:\Users\Jhim\Downloads\
        *EXTRA Dir        -1    C:\Users\Jhim\Favorites\
        *EXTRA Dir        -1    C:\Users\Jhim\IntelGraphicsProfiles\
        *EXTRA Dir        -1    C:\Users\Jhim\ITEfolder1\
        *EXTRA Dir        -1    C:\Users\Jhim\ITEfolder2\
        *EXTRA Dir        -1    C:\Users\Jhim\ITEfolder3\
        *EXTRA Dir        -1    C:\Users\Jhim\Links\
        *EXTRA Dir        -1    C:\Users\Jhim\Local Settings\
        *EXTRA Dir        -1    C:\Users\Jhim\Music\
        *EXTRA Dir        -1    C:\Users\Jhim\My Documents\
        *EXTRA Dir        -1    C:\Users\Jhim\NetHood\
        *EXTRA Dir        -1    C:\Users\Jhim\OneDrive\
        *EXTRA Dir        -1    C:\Users\Jhim\Pictures\
        *EXTRA Dir        -1    C:\Users\Jhim\PrintHood\
        *EXTRA Dir        -1    C:\Users\Jhim\Recent\
        *EXTRA Dir        -1    C:\Users\Jhim\Saved Games\
        *EXTRA Dir        -1    C:\Users\Jhim\Searches\
        *EXTRA Dir        -1    C:\Users\Jhim\SendTo\
        *EXTRA Dir        -1    C:\Users\Jhim\Start Menu\
        *EXTRA Dir        -1    C:\Users\Jhim\Templates\
        *EXTRA Dir        -1    C:\Users\Jhim\Tracing\
        *EXTRA Dir        -1    C:\Users\Jhim\Videos\
        *EXTRA Dir        -1    C:\Users\Jhim\WPS Cloud\
        *EXTRA Dir        -1    C:\Users\Jhim\WPS Cloud Files\
        *EXTRA Dir        -1    C:\Users\Jhim\WPSDrive\
          *EXTRA File               174        .packettracer
          *EXTRA File            24.2 m        NTUSER.DAT
          *EXTRA File             6.2 m        ntuser.dat.LOG1
```

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```
        *EXTRA File            1.8 m        ntuser.dat.LOG2
        *EXTRA File                0        NTUSER.DAT{b4519911-defd-11ef-a390-c8b9b1a3107a}.TxR.0.regtrans-ms
        *EXTRA File                0        NTUSER.DAT{b4519911-defd-11ef-a390-c8b9b1a3107a}.TxR.0.regtrans-ms.cnpf
        *EXTRA File                0        NTUSER.DAT{b4519911-defd-11ef-a390-c8b9b1a3107a}.TxR.1.regtrans-ms
        *EXTRA File                0        NTUSER.DAT{b4519911-defd-11ef-a390-c8b9b1a3107a}.TxR.1.regtrans-ms.cnpf
        *EXTRA File                0        NTUSER.DAT{b4519911-defd-11ef-a390-c8b9b1a3107a}.TxR.2.regtrans-ms
        *EXTRA File                0        NTUSER.DAT{b4519911-defd-11ef-a390-c8b9b1a3107a}.TxR.2.regtrans-ms.cnpf
        *EXTRA File                0        NTUSER.DAT{b4519911-defd-11ef-a390-c8b9b1a3107a}.TxR.blf
        *EXTRA File                0        NTUSER.DAT{b4519911-defd-11ef-a390-c8b9b1a3107a}.TxR.blf.cnpf
        *EXTRA File            73728        NTUSER.DAT{b4519912-defd-11ef-a390-c8b9b1a3107a}.TM.blf
        *EXTRA File             4096        NTUSER.DAT{b4519912-defd-11ef-a390-c8b9b1a3107a}.TM.blf.cnpf
        *EXTRA File           561152        NTUSER.DAT{b4519912-defd-11ef-a390-c8b9b1a3107a}.TMContainer00000000000000000001.regtrans-ms
        *EXTRA File             4096        NTUSER.DAT{b4519912-defd-11ef-a390-c8b9b1a3107a}.TMContainer00000000000000000001.regtrans-ms.cnpf
        *EXTRA File           561152        NTUSER.DAT{b4519912-defd-11ef-a390-c8b9b1a3107a}.TMContainer00000000000000000002.regtrans-ms
        *EXTRA File             4096        NTUSER.DAT{b4519912-defd-11ef-a390-c8b9b1a3107a}.TMContainer00000000000000000002.regtrans-ms.cnpf
        *EXTRA File               20        ntuser.ini
        New Dir         0    C:\Users\Jhim\ITEfolder3\ITEfolder4\ITEfolder5\

------------------------------------------------------------------------------

               Total    Copied   Skipped  Mismatch    FAILED    Extras
    Dirs :         2         1         1         0         0        36
   Files :         0         0         0         0         0        19
   Bytes :         0         0         0         0         0   33.49 m
   Times :   0:00:00   0:00:00                       0:00:00   0:00:00
   Ended : Tuesday, 6 October 2026 8:23:19 pm
```

Note the information provided by the command during the copying process.

h. Navigate to **C:\Users\ITEUser** to verify that the folder ITEfolder5 is copied.

```
C:\Users\Jhim>dir
 Volume in drive C is OS
 Volume Serial Number is A448-0424

 Directory of C:\Users\Jhim

06/10/2026  08:23 pm    <DIR>          .
30/01/2025  07:31 pm    <DIR>          ..
30/11/2024  01:25 pm    <DIR>          .android
08/01/2024  10:14 am    <DIR>          .arduinoIDE
05/02/2024  08:07 am    <DIR>          .idlerc
25/01/2026  11:18 pm    <DIR>          .mgltools
01/09/2026  11:43 am               174 .packettracer
23/02/2024  10:14 am    <DIR>          3D Objects
01/09/2026  01:07 pm    <DIR>          Cisco Packet Tracer 9.0.1
30/01/2025  07:38 pm    <DIR>          Contacts
04/11/2025  01:59 pm    <DIR>          Documents
06/10/2026  07:07 pm    <DIR>          Downloads
30/01/2025  07:38 pm    <DIR>          Favorites
06/10/2026  08:20 pm    <DIR>          ITEfolder1
06/10/2026  07:47 pm    <DIR>          ITEfolder2
06/10/2026  07:50 pm    <DIR>          ITEfolder3
06/10/2026  07:18 pm    <DIR>          ITEfolder5
30/01/2025  07:38 pm    <DIR>          Links
30/01/2025  07:38 pm    <DIR>          Music
06/10/2026  06:50 pm    <DIR>          OneDrive
08/04/2022  08:45 am    <DIR>          Pictures
30/01/2025  07:38 pm    <DIR>          Saved Games
30/01/2025  07:38 pm    <DIR>          Searches
18/03/2022  11:00 am    <DIR>          Tracing
13/07/2026  12:53 am    <DIR>          Videos
15/07/2026  10:08 pm    <DIR>          WPS Cloud
22/04/2026  10:36 pm    <DIR>          WPSDrive
               1 File(s)            174 bytes
              26 Dir(s)  88,407,552,000 bytes free

C:\Users\Jhim>
```

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Step 6: Delete directories.

In this step, you will delete an empty and a non-empty directory using the **rd** command.

a. Use the **rd ITEfolder2** to delete the empty directory and verify that the directory was deleted.

```
C:\Users\ITEUser> rd ITEfolder2
```

```
C:\Users\Jhim>rd ITEfolder2

C:\Users\Jhim>dir
 Volume in drive C is OS
 Volume Serial Number is A448-0424

 Directory of C:\Users\Jhim

06/10/2026  08:33 pm    <DIR>          .
30/01/2025  07:31 pm    <DIR>          ..
30/11/2024  01:25 pm    <DIR>          .android
08/01/2024  10:14 am    <DIR>          .arduinoIDE
05/02/2024  08:07 am    <DIR>          .idlerc
25/01/2026  11:18 pm    <DIR>          .mgltools
01/09/2026  11:43 am               174 .packettracer
23/02/2024  10:14 am    <DIR>          3D Objects
01/09/2026  01:07 pm    <DIR>          Cisco Packet Tracer 9.0.1
30/01/2025  07:38 pm    <DIR>          Contacts
04/11/2025  01:59 pm    <DIR>          Documents
06/10/2026  07:07 pm    <DIR>          Downloads
30/01/2025  07:38 pm    <DIR>          Favorites
06/10/2026  08:20 pm    <DIR>          ITEfolder1
06/10/2026  07:50 pm    <DIR>          ITEfolder3
06/10/2026  07:18 pm    <DIR>          ITEfolder5
30/01/2025  07:38 pm    <DIR>          Links
30/01/2025  07:38 pm    <DIR>          Music
06/10/2026  06:50 pm    <DIR>          OneDrive
08/04/2022  08:45 am    <DIR>          Pictures
30/01/2025  07:38 pm    <DIR>          Saved Games
30/01/2025  07:38 pm    <DIR>          Searches
18/03/2022  11:00 am    <DIR>          Tracing
13/07/2026  12:53 am    <DIR>          Videos
15/07/2026  10:08 pm    <DIR>          WPS Cloud
22/04/2026  10:36 pm    <DIR>          WPSDrive
               1 File(s)            174 bytes
              25 Dir(s)  88,399,806,464 bytes free
```

b. Navigate to the **C:\Users\ITEUser\ITEfolder3** directory.

c. Use the **rd ITEfolder1** to delete the directory. Verify the directory removal using the **dir** command.

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```
C:\Users\Jhim>cd ITEfolder3

C:\Users\Jhim\ITEfolder3>rd ITEfolder1
The system cannot find the file specified.

C:\Users\Jhim\ITEfolder3>dir
 Volume in drive C is OS
 Volume Serial Number is A448-0424

 Directory of C:\Users\Jhim\ITEfolder3

06/10/2026  07:50 pm    <DIR>          .
06/10/2026  08:33 pm    <DIR>          ..
06/10/2026  07:36 pm                20 file1.txt
06/10/2026  07:37 pm                20 file2.txt
06/10/2026  07:18 pm    <DIR>          ITEfolder4
               2 File(s)             40 bytes
               3 Dir(s)  88,399,331,328 bytes free
```

Were you able to delete the directory? Explain.

No, I was not able to delete the directory. The rd command only removes empty directories, and ITEfolder1 still contains files and subfolders.

d. Use **rd /?** command to determine the switch that allows the deletion of a non-empty directory.

```
c:\Users\ITEUser\ITEfolder3> rd /S ITEfolder1
ITEfolder4, Are you sure (Y/N)? y
```

```
C:\Users\Jhim\ITEfolder3>rd /S ITEfolder1
ITEfolder1, Are you sure (Y/N)? Y
The system cannot find the file specified.
```

e. Use the appropriate commands to delete all the text files and folders that you have created in this lab. Type **exit** to close the command prompt window.

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```
C:\Users\Jhim\ITEfolder3>cd C:\Users\jhim

C:\Users\Jhim> rd /S /Q ITEfolder1

C:\Users\Jhim> rd /S /Q ITEfolder3

C:\Users\Jhim> rd /S /Q ITEfolder5

C:\Users\Jhim>dir
 Volume in drive C is OS
 Volume Serial Number is A448-0424

 Directory of C:\Users\Jhim

06/10/2026  08:38 pm    <DIR>          .
30/01/2025  07:31 pm    <DIR>          ..
30/11/2024  01:25 pm    <DIR>          .android
08/01/2024  10:14 am    <DIR>          .arduinoIDE
05/02/2024  08:07 am    <DIR>          .idlerc
25/01/2026  11:18 pm    <DIR>          .mgltools
01/09/2026  11:43 am               174 .packettracer
23/02/2024  10:14 am    <DIR>          3D Objects
01/09/2026  01:07 pm    <DIR>          Cisco Packet Tracer 9.0.1
30/01/2025  07:38 pm    <DIR>          Contacts
04/11/2025  01:59 pm    <DIR>          Documents
06/10/2026  07:07 pm    <DIR>          Downloads
30/01/2025  07:38 pm    <DIR>          Favorites
30/01/2025  07:38 pm    <DIR>          Links
30/01/2025  07:38 pm    <DIR>          Music
06/10/2026  06:50 pm    <DIR>          OneDrive
08/04/2022  08:45 am    <DIR>          Pictures
30/01/2025  07:38 pm    <DIR>          Saved Games
30/01/2025  07:38 pm    <DIR>          Searches
18/03/2022  11:00 am    <DIR>          Tracing
13/07/2026  12:53 am    <DIR>          Videos
15/07/2026  10:08 pm    <DIR>          WPS Cloud
22/04/2026  10:36 pm    <DIR>          WPSDrive
               1 File(s)            174 bytes
              22 Dir(s)  88,393,687,040 bytes free

C:\Users\Jhim>
```

Reflection Question

What are the advantages of using CLI vs. GUI?

The command-line interface (CLI) is generally faster and more efficient for repetitive or large-scale tasks. A single command with a wildcard, such as move file* or del doc2*, can act on dozens of files at once, whereas a GUI would require selecting each item manually. CLI commands can also be saved in batch files and scripts, which makes tasks easy to automate, repeat, or schedule. In addition, the CLI uses very little system memory and can be used over remote connections on servers that have no graphical interface at all.

The CLI also gives users more precision and control than a GUI. Switches and options, such as /E in xcopy or /S in rd, along with tools like robocopy, offer capabilities such as mirroring directories and resuming interrupted copies that most graphical tools do not expose. Commands can be combined and

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redirected, as with echo ... > file.txt, to build powerful workflows in a single line. The trade-off is a steeper learning curve, since users must memorize exact syntax, and mistakes like rd /S are permanent because there is no Recycle Bin to recover from.

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